

CHILLICOTHE, MO, USA

WMO: 724464

Lat: 39.823N

Lon: 93.579W

Elev: 769

StdP: 14.29

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 53916

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	6.1	-8.2	3.6	1.7	-4.0	4.6	8.0	19.9	30.2	18.2	33.7	4.0	310	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	95.1	76.8	92.2	76.3	89.8	75.4	80.1	90.5	78.4	88.3	76.9	86.4	7.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.1	145.3	87.2	75.4	136.8	85.1	73.9	130.2	83.0	44.2	89.9	42.5	88.4	40.8	86.2	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.8	15.9	13.3	-8.2	98.8	5.3	3.3	-12.0	101.2	-15.1	103.2	-18.1	105.0	-22.0	107.5		
			-8.8	82.2	5.1	1.9	-12.4	83.6	-15.4	84.7	-18.3	85.8	-22.0	87.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	27.9	30.7	44.2	54.7	64.4	74.1	77.9	76.0	68.1	55.4	43.8	32.8
	DBStd	19.73	12.08	12.00	11.34	9.57	8.35	5.95	5.57	5.49	7.65	9.53	9.99	10.49
	HDD50	2371	687	547	251	59	5	0	0	0	51	234	537	
	HDD65	5225	1151	960	649	328	116	6	0	1	53	324	638	999
	CDD50	3936	2	7	70	199	451	722	866	807	542	220	47	3
	CDD65	1315	0	0	3	18	98	278	402	343	145	27	1	0
	CDH74	13187	0	0	28	194	877	2760	4405	3326	1342	251	4	0
	CDH80	5276	0	0	1	32	235	1065	2010	1419	455	59	0	0
Wind	WSAvg	5.7	6.5	6.6	7.4	7.3	5.6	5.1	4.3	3.9	4.2	5.2	6.1	6.1
Precipitation	PrecAvg	39.4	1.2	1.9	2.6	3.9	5.3	5.0	4.5	4.3	4.2	2.7	2.2	1.7
	PrecMax	59.9	2.9	5.0	6.1	6.1	12.4	9.5	15.7	11.7	12.4	7.6	6.5	4.3
	PrecMin	25.8	0.2	0.3	0.4	1.1	1.1	1.1	0.9	0.0	0.7	0.2	0.2	0.1
	PrecStd	8.3	0.7	1.1	1.5	1.5	2.8	2.3	3.1	2.4	2.8	1.8	1.5	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.9	70.4	77.6	84.0	89.8	95.2	100.1	99.6	92.5	85.9	73.6	63.6
		MCWB	51.9	53.6	63.3	66.7	72.1	76.7	75.1	75.0	75.8	71.5	60.1	56.3
	2%	DB	55.3	61.5	72.9	79.3	85.5	91.5	95.7	94.5	88.8	80.7	69.6	56.5
		MCWB	46.9	51.4	59.9	63.3	70.3	75.7	77.4	76.1	74.1	66.8	59.2	49.8
	5%	DB	50.2	54.1	68.1	75.3	82.3	89.1	92.8	91.0	85.5	75.9	64.3	52.1
		MCWB	43.7	46.3	57.1	61.6	68.5	74.6	77.2	75.8	72.3	64.2	54.6	46.5
	10%	DB	44.7	48.3	63.1	71.8	79.3	86.2	90.0	87.8	81.7	71.8	59.9	47.7
		MCWB	39.3	42.0	54.4	59.8	66.7	72.9	76.2	74.9	69.6	62.3	52.0	42.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	58.0	66.0	68.8	75.2	80.0	82.3	82.1	77.6	73.4	63.8	58.6
		MCDB	58.7	65.2	75.0	81.1	85.4	90.4	92.2	92.8	88.8	82.0	70.5	62.3
	2%	WB	48.0	51.9	62.4	66.3	72.7	78.0	80.5	79.7	75.7	70.4	59.9	51.6
		MCDB	53.3	59.9	69.4	76.2	82.6	88.0	90.8	90.1	85.4	77.5	67.1	54.3
	5%	WB	43.7	47.0	59.3	63.9	70.5	76.2	79.0	77.7	74.2	66.9	56.6	46.7
		MCDB	50.1	53.4	65.8	73.0	79.7	86.1	89.2	87.4	82.2	73.1	63.0	51.4
	10%	WB	39.2	41.9	54.6	61.0	68.6	74.6	77.4	76.0	72.2	63.2	52.8	42.8
		MCDB	44.6	48.1	62.4	69.9	76.9	84.0	86.8	84.8	79.2	70.8	58.7	47.1
Mean Daily Temperature Range	5% DB	MDBR	18.3	19.6	20.4	21.3	20.3	19.4	19.8	20.1	21.9	21.3	19.8	17.0
		MCDBR	26.1	28.7	26.9	26.3	23.2	21.6	22.3	23.9	23.4	25.5	26.0	23.5
		MCWBR	18.6	19.5	16.0	14.9	11.3	9.6	8.8	9.7	10.4	13.3	16.9	17.7
	5% WB	MCDBR	24.9	26.9	22.7	21.9	20.1	19.1	18.9	20.0	19.9	20.1	22.6	22.0
		MCWBR	18.6	19.7	14.8	14.0	10.8	9.8	9.4	10.0	9.7	13.0	16.7	18.4
Clear-Sky Solar Irradiance	taub	0.298	0.315	0.340	0.392	0.427	0.428	0.441	0.442	0.389	0.345	0.303	0.296	
	taud	2.437	2.388	2.393	2.264	2.232	2.262	2.225	2.242	2.341	2.459	2.519	2.467	
	Ebn at Noon	277	287	290	280	271	270	265	262	270	272	274	268	
	Edh at Noon	25	31	34	42	44	43	44	42	36	29	23	23	
All-Sky Solar Radiation	RadAvg	643	917	1197	1542	1803	2011	2018	1764	1511	1040	717	562	
	RadStd	69	83	104	106	153	172	132	89	92	115	68	59	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page